

The Social Radars

S3: Kyle Vogt, Founder of Cruise Automation

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Jessica: I'm Jessica Livingston, and Carolynn Levy and I are The Social Radars. In this podcast we talk to some of the most successful founders in Silicon Valley about how they did it. Carolynn and I have been working together to help thousands of startups at Y Combinator for almost 20 years. Come be a fly on the wall as we talk to founders and learn their true stories. Carolynn, we have a very exciting guest today. I'm so excited to welcome Kyle Vogt on the show today. Hi, Kyle.

Carolynn: Hi, Kyle.

Kyle: Thanks for having me.

Jessica: You're welcome. Kyle is one of the founders of Justin.tv. He then went on to be one of the founders of Twitch, and then he started Cruise, which is the self-driving car technology, which went through Y Combinator in winter 2014 and was acquired by General Motors in 2016. That's the background. Welcome.

Kyle: Yeah, this is going to be fun. Looking forward to it.

Jessica: We have a lot to get through, I'm just going to tell you right off the bat, because we've known you for so long. You were how old?

Kyle: I think I met Justin Emmett and quickly thereafter, Paul, I must have been 20, 21, somewhere around there.

Jessica: Yeah, I think you were 20. I want to go back to when you were in college. You were at MIT, and how did you get involved with Justin.tv? Tell us the story?

Kyle: Well, at MIT, you know the thing I loved about being there was the people. Because there was always someone willing to hack on an interesting project. And so I spent less time in classes and more time building things. We found a safe in the basement of some building, so we spent three months building a safe-cracking robot just to open it. We just really-

Jessica: Wait, what was inside?

Kyle: It was nothing inside.

Jessica: No.

Kyle: I loved doing it. And there'd be these engineering contests and there was the DARPA Grand Challenge, which is the autonomous vehicle race sponsored by the government. There's tons of cool stuff going on. I loved building. I kind of knew since high school that I'd wanted to start a company. My plan was to graduate and then probably go off and either join a big tech company for a while, then maybe start a company, but I didn't really have a lot of clarity on that. But what I was doing at MIT was monitoring this mailing list. I joined the entrepreneurship club, I was trying to meet people and there's this computer science mailing list run by an administrator named Anne Hunter, who I learned just retired a few years ago. But it was like a jobs list. Anyone could post to it and would go out to all the computer science and electrical engineering students.

And what ended up happening, it was kind of taken over mostly by Harvard MBAs and sometimes Yales who they'd have this grand business idea. They'd write up their proposal for how they're going to change the world of technology and they just needed to find some like MIT software person to go off and build it for them. And so most of these emails that came through were very pretentious and off-putting. Then I saw the one from Justin and Emmett. They were looking for someone to build a live-streaming camera, and I Googled them and saw that they had just sold their last company Kiko on eBay for \$250,000. And I thought this is a fortune. And I assume they were driving around in Ferraris and Lamborghinis, so they were in Boston and I agreed to meet with them.

They were not in a Ferrari, maybe a Kia. What I thought was interesting is I could probably learn from these folks, because they had built a company before they had sold it. And I thought their idea, I was kind of skeptical, but I thought it was an interesting piece of technology to build, this live-streaming camera to stream this person's life to the internet. And so I signed up to build this first version of the camera, because I thought it was a cool technical challenge and maybe it'll turn into something. And I did. And then of course, flew out to San Francisco later on and got sucked into the world of actually building that company.

Jessica: I need to interrupt you, because you said you knew in high school that you thought you'd want to start a company, weren't you from Kansas City? How did you get this bug in high school? What were you exposed to there?

Kyle: Well, the two things I was interested in were robotics and that I think, I had toys and things, Legos and all that stuff that I liked building. But the thing that really got me excited was seeing the BattleBots show, the TV show on Comedy Central where you have radio-controlled fighting robots.

Jessica: I've seen that.

Kyle: And so I spent my summers, I got a job at RadioShack, paid basically minimum wage, because they gave me a discount on parts. And so I was working using the money to build this battle bot in my basement and then eventually went out even though I loved robotics and so knew that MIT was the place to go for that. But I was also trying, I built these circuit boards and things for these robots and I'd be like, "Oh, this is a motor controller. I should try to sell this online, or I should start a little business like building

parts and components for robots." And something about that seemed really appealing, but I didn't go all in on trying to start a company then, it was kind of on the back burner at that point.

Jessica: Okay. But you were really interested in it? You sound like Sheldon Cooper with RadioShack. His obsession with RadioShack.

Kyle: RadioShack was great back in its day. It's the only place you could go to find a 300 ohm resistor in the middle of the day to finish a project.

Jessica: Carolynn and I did not have those projects. We were in high school.

Carolynn: No, but I definitely remember RadioShack. I mean, I'm sorry, I bought batteries there or something.

Jessica: Okay. So you go to MIT because you say that's where all the smart engineers go. You spend time trying to crack the safe in the basement that you found, not doing schoolwork. Were you thinking like, "Oh, I want to drop out and join something?"

Kyle: Well, I remember at MIT being, so this was 2004, 2006 time period, and I remember feeling a little frustrated almost. I wouldn't say bait and switch, but a little frustrated, because I liked building things. But all the classes I would go to were very fundamental, pure mathematics, so far from applied engineering that it felt like if I wanted to do any of that fun stuff, which in my world was actually building real contraptions projects, writing code, I'd have to do it mostly on my own. Like side projects or in contests or other things. And so that was a little weird. And then I felt like as I watched a lot of my classmates, especially the older ones go off, they were all getting jobs in finance or consulting. That was the thing to do.

Jessica: Oh, that's depressing.

Kyle: And so it was a competition to get into one of the elite New York trading groups that was seen as the best. And then this is also the time Google started putting up their billboards, like, "Solve this equation on MIT's campus," which was pretty cool. And then people started going to Apple and Google as the elite job. There was no notion of doing startups. And when I looked at the jobs that you would get if you went to Apple or Google, again, you're not building stuff. You're building a tiny, tiny piece of something or testing some component of a much larger system, and it didn't seem fun. I was a little uneasy with the trajectory that I seemed to be on if I continued forward at MIT. And so I think when this alternative popped up, as wacky as it was, it was just pulling me because the default path wasn't a great path.

Jessica: Got it. Okay, that makes sense. You mentioned the DARPA Grand Challenge, which had self-driving cars, right?

Kyle: Yeah. I look back on that. One of the best uses of government money in history, for a million dollar prize they convinced dozens of academic institutions to throw all their best

people and grad students at undergrad spending an infinite amount of time trying to build these robots. And the task was to drive about 50 miles through the desert in this race. And it was a great use of government funding to kickstart this industry that had been sleepy and fragmented and turn it into something exciting. And from that, the people who competed in those teams in those initial DARPA grand Challenge went off to go and form every single AV company that exists today over a decade later. But quite literally, if you look at the founders of every single major multi-billion dollar AV company, they almost all came from this initial competition, which is cool.

Carolynn: That is really cool.

Jessica: I know, I had no idea. Oh, wow.

Carolynn: Me neither.

Jessica: Okay, so we're going to put that to the side for a second, but we're going to remember that you participated in that. You go off, you move to California, join the Justin.tv guys, so that's Emmett, Justin and Michael Seibel and you co-founders of this crazy new technology, which was incredibly difficult, because back then Justin was on 24/7, and so you had to figure out how to keep him connected to the internet.

Kyle: Yeah, it sounds trivial today, but we were carrying around flip phones. And they would have Edge or 3G on them, which is a predecessor to 4G and 5G kind of where we are now. And so the notion of, this was when you could send grainy images through SMS to each other, that was the state of using your phone to do multimedia. The notion of doing a live stream to a website from a mobile device period was pretty interesting. But then doing it as you're walking around a city, you're walking into bars and basements where the walls and the buildings limit the reception, it was a big challenge. And so first we got it working at all.

And then the big next challenge, at least from my point of view, was to add a whole bunch of different cellular modems. You'd be connected to ATT, Verizon, T-Mobile simultaneously and use all of them, like whichever one would give you the most data, you'd send more data through it. And the end goal was just to have a continuous clear feed of Justin, even as he's walking around through crowded areas and in out of bars and getting in and out of cars. And so it was fun. It was a fun technical challenge. No one had really done that. And it mostly worked after a few tries, enough that I think you could follow along in Justin's life. And I think that was the point.

Jessica: Didn't he have a backpack that carried all the equipment. And wasn't it super heavy and you had to figure out how to lighten it?

Kyle: Yeah, I mean, so the initial thing was a camera, a small computer that would compress the video and split it across these different modems, and then a bunch of batteries. Because this thing, you could use your phone to record or stream a video, but if you did that continuously, you'd burn through the battery pretty quick. And so we actually had 10 pounds of batteries in this backpack so that it could go up to I think about 12 hours

without swapping batteries. And we had to make sure that even when he's swapping batteries, you can't just turn the thing off to recharge it. We had to make it hot swappable. We had batteries on standby and he'd plug in a new one, flip a switch to engage that battery and then unplug the old one to go charge it so it would never shut off basically.

Jessica: I mean, Carolyn and I are both speechless, because it just seems so hard. I mean, do you think that because you knew about hardware and software, you were especially equipped to create this stuff?

Kyle: Well, look, I mean, having worked with a bunch of brilliant engineers for years, looking back on this, I know a lot of people who could have done this job. You just need the right person. And I think Justin and Emmett's challenge quite frankly was finding someone with that skill set that would be willing to work with them on this crazy project for basically no money. That was actually the challenge, not, I don't think the limiting reagent was good engineers, there were plenty of people who could do that. But it was really finding one who was willing to sign up for their crazy mission.

Jessica: They did have a crazy mission. We talked in the Emmett interview. I remember saying, "That's the stupidest idea I've ever heard." And thank God they didn't listen to me. But what was it about the team, besides the fact that all four of you believed that this could work and wanted to spend your time doing this crazy idea? Was there anything else special about the four of you that contributed to the success?

Kyle: They were super smart, first of all, that became clear. They had had some experience and so I figured personally there'd be an opportunity to learn. The technical thing we were working on was fun, and hanging out with that group of people was really fun. To me, it was really almost in the beginning of Justin.tv was really like the extension of the college dorm or living group life, because there were four of us and usually one or two visiting guests or visitors in a two-bedroom apartment in North Beach. And so, it was a familiar environment to me, I guess coming from college, but it was just a lot of fun. There was this undertone of we're going to work 24/7 and it is extremely important that this company succeed. But when you're around people 24/7, there's plenty of opportunities to mess around and have fun. And it ended up being a really ideal work-life hybrid setup for a bunch of young 20-somethings.

Jessica: Were you in Crystal Towers, this Y-Scraper?

Kyle: The Y-Scraper? That's right.

Jessica: There were a lot of other startup founders in the building?

Kyle: We were on floor six, I think Wheely was also on floor six, and then Reddit was on floor nine. Basically, all the tech companies of that era were in that one building, which is really cool.

Jessica: Would you guys help each other at all? I mean, not obviously with big problems, but with little things? Did you have this little community there?

Kyle: Yeah, people would work all day and then maybe around nine or 10 o'clock, they did want to go grab a drink or we'd just hang out and usually there'd be someone venting about the problem of the day and just casually talking about it. We didn't have any formal, "Let's get the engineers in a room and sit down and work through our problems." It was more just osmosis from hanging out and talking about what we were working on. We all found each other's problems really interesting. And so I think there was a fair amount of cross-pollination of ideas that was really helpful, almost like having a much larger team to lean on when you get stuck, or a bunch of mentors. But yeah, it was a really great environment because not only was there some technical exchange or business exchange or fundraising advice, but also there were a bunch of people living in the exact same way. Meaning they were working 24/7 and then willing to hang out and blow off some steam late at night or at two or three in the morning once they finish working on something.

And so that made the startup existence, which can be really lonely, I think a lot more social and tolerable, especially when you're going through all those ups and downs as everyone does in the early stages of a business.

Carolynn: Did you spend most of your time fixing things that broke or did you get to spend a lot of time actually evolving the technology that Justin was wearing?

Kyle: Well, there was a time period, I feel like it was about six months where the website and the viewership for Justin.tv was growing exponentially. And I had never run a whole bunch of servers or built a live video stream or had any site reliability engineering or any knowledge to keep all these systems up and running. There's a lot of firefighting and figuring out how to make these things scale. Running video streaming in 2006, 2007 actually was hard. Amazon Web Services had just come out. We were one of the first users, because we thought it was a pain to keep buying servers and physically driving to the data server and racking them up, which is basically what you had to do back then. A lot of time on that.

And so, my job was to basically keep the streaming system up and running so that the show could run, and it was 24/7, so it was like there's no downtime, and if you haven't go out for even a minute, there's hundreds or thousands of people who instantly start complaining and texting and emailing, where it's like, "We're down for 10 minutes, people don't notice, they just go back later and hit refresh." But not one of these.

Jessica: Did you sleep? Did you have a weird sleep schedule?

Kyle: Well, that was kind of the issue. I would stay up really late because then everyone else was asleep and I could work on the next generation of the camera or the new products and then ideally would have wanted to sleep in very late. But in practice, Michael Seibel was up at 7:00 a.m. making phone calls and we were basically living in the same room. And so I was up at 7:00 a.m., and so basically I just didn't sleep for several months. That

was very hard on my body physically and very stressful, but I chalked it up to being in my early 20s is the only way I could get through that.

Carolynn: I was just going to say, it's a young person's game sometimes.

Jessica: Yeah. Oh, wow. Sorry, my mind is carried back. I remember your apartment. I remember visiting you guys. It did feel like a dormitory. And do you remember, we didn't bring this up with Emmett, Carolynn, but do you remember when the police came to your apartment and Justin, they're like, "Put your hands up," because they said there was a knife attack going on?

Kyle: Yeah.

Jessica: What happened?

Kyle: Well, so I think this concept of swatting, which is now well known in Twitch in other worlds where you basically prank call the cops on someone on a live stream, I was indented in real time in Justin.tv, because once people figured out, some kid somewhere sitting in his mama's basement probably figured out that, "I can watch this video stream and I can interact with it. Wherever Justin goes, I can call the restaurant and ask to talk to him or something like that." And that escalated very quickly where there was calling firetrucks to the building when we were having poker nights or all that kind of stuff. And one day, yes, there was an incident where someone called 911 and said there was a stabbing in apartment 605 in Crystal Towers.

And so the police go there apparently, and my desk, actually where I was building all the hardware for these cameras, was the closest one to the door, so I was the informal door answerer for the apartment. And so there's a knock on the door and it says, "SF police, open up." It sounded a lot like one of my friends who lived in the building. So, I swung open the door ready to be like, "Hey, man, good to see you." And there's a police officer with the gun drawn, pointed probably 18 inches from my face.

Jessica: Oh, my God.

Kyle: That was very real and very, very disturbing. I'm lucky to have survived that, because swinging the door open abruptly is not a good way to respond to something like that. But they came in three or four police officers and looked around the room and they said a bunch of, I think Justin wasn't even wearing a shirt. It's like three guys sitting on a couch on one side and me sitting at a desk full of wires and everything, and they got out of there pretty quickly, but it was the first, well, one of many incidents we had where people were just messing around with Justin while he was on camera.

Jessica: And the whole thing was recorded?

Kyle: It was on [inaudible 00:18:25] the next day, which was the big news site of the time.

Jessica: I shouldn't have laughed, because it is quite frightening, but in a way I do remember that that incident did make the headlines.

Kyle: Sometimes you get in the zeitgeist and then suddenly anything that happens becomes news. And that happened for a while with Justin. I remember, he would go on a date and TechCrunch would write an article about it. "His date's tonight at 7:00 p.m., tune in." It's a very strange moment in time when I think media, especially tech media, can get fixated on something. You see it all the time. I don't know, the big open AI board situation, every minute update became a new headline for some period of time, and that's what happened for a while with Justin.tv.

Jessica: So after Justin.tv it sort of morphed into Twitch?

Kyle: Yeah, I'm sure Emmett told that story pretty well, but-

Jessica: He did. But what are some of your memories of working at Twitch early on?

Kyle: Well, I remember the moment when we were trying to figure out what to do with Justin.tv, and one of the other things I remember about working with Michael, Justin and Emmett, they're all from Yale and I feel like there's a culture there of perpetual arguing, almost the sport of arguing with each other. And so they often didn't agree, and it seemed like they just would argue for hours and they were trying to solve a problem, but mostly just enjoying batting with each other on things. And so we were trying to figure out is mobile the future? Is gaming the future? Is there some way to make Justin.tv as it exists, work? It was just a deadlock. And so that's why Michael went off and led the social cam initiative. Emmett went off and led the Twitch initiative and I was doing the engineering across all of them at the time, just trying to support all of these things.

It's pretty interesting. I remember one thing that stood out that I still respect Emmett for is when we were trying to figure out how to make this site great for gamers, he did things that most engineers aren't willing to do, which is pick up the phone and call the users. And so he made a list of 100 people that were users or the top users on our site or other sites and called them up and just listened and said, "What would make streaming video gaming on our site awesome?" Basically he made a list and that became our product roadmap. And then he would follow up and call people and be like, "Hey, we built this feature. You said you wanted, will you come stream on our website now?" And after a while it worked. It was a really low-tech gritty method to bootstrap the site.

Jessica: Hold on. Am I remembering now correctly that people were using Justin.tv to show them playing video games? That was a popular thing?

Kyle: Yeah, we had categories. We had 20 different verticals of content on the site, and video gaming was one of them. And out of all of them that we looked at, it was the only one that was content that didn't have any copyright issues associated with it. It wasn't like people streaming cable TV or something, and it was growing organically week over week without us doing anything. And so Emmett saw that, we all could see and agree that, "Hey, this thing, it's small in terms of numbers, but it's growing every week. So maybe if

we keep building and focus on things for just those users, it will grow really fast and turn into something."

Jessica: Okay, so Emmett called, he actually didn't talk about that. He talked to the users, he got their feedback, and then you guys launched Twitch?

Kyle: I think it was more of a continuous thing, like building all the things that they wanted to one after the other. And at some point in time it made sense to launch Twitch, which is really a clone of the Justin.tv code base just set up on a new website, with all the other categories shut off. And that was how it initially started. And from there the traffic started to grow there. The traffic on Justin.tv was slowly declining and it became clear what we needed to do.

Jessica: Okay. How long were you at Twitch for?

Kyle: I don't remember exactly when Justin.tv formally became Twitch, because to me it was kind of a spectrum over time. But I was at the Justin.tv/Twitch combo for seven or eight years and actually left, I think about a year before the Amazon acquisition.

Jessica: Oh, wow. Now, did you leave because you wanted to start Cruise? Why did you leave?

Kyle: Well, the thing that brought me into Justin.tv early on, one of the most attractive things was the technical challenges. And after it became a live-streaming platform, obviously, building cameras was no longer important because people were just sending data from their laptops and webcams for video game consoles. The next big challenge became figuring out how to do really cheap live video delivery globally. Because in the early days we were only making money on banner ads on the top of each page, and someone would load the web page, we'd make a fraction of a penny on them seeing that one banner ad, and then they'd sit there and consume an hour worth of bandwidth. And so economically, this was not good.

And so, I spent a ton of time figuring out how do we, from first principles, get the cheapest way to get bits from people's webcam somewhere in the world to everyone else's screen, and then spent a lot of time building out this global live delivery network. We did some really innovative things that were optimized just for live-streaming traffic, which is very different than serving just images and static videos on the web. Basically, the key constraint is you can never let any, the network, it is a series of tubes in a way, but any connection between any router or server, when you send a piece of data from your laptop to a server, someone's website, it will hop through three or four different telecom networks and maybe half a dozen or a dozen routers. And the problem with live video is if any one of those links between you and that server gets saturated and meets capacity, you'll have a buffering, choppy live video stream.

And the problem is most of these networks, especially the ones who are trying to run things economically, let their pipes fill up and have packet loss and sort of bad performance all the time. So, we had to build a system to dynamically identify those

hotspots and route the traffic around it so that you would always get a smooth video stream and not buffering.

Jessica: Oh, man.

Kyle: That was a technical problem, but eventually we solved that, and I'm not really a gamer, so the Twitch thing wasn't pulling me towards it because of the intrinsic excitement about the content. And the big technical challenges were mostly solved, at least from my perspective, that the 80% part of it was solved, and so I was ready for something else. The last six months before I left, I knew I was about there and was thinking about other things I might want to do next. I tried, at the time Gmail search was really slow weirdly, and so I was like, "Okay, I'm going to build a Gmail competitor that's faster." So search is always instantaneous.

Carolynn: I really wish you'd done that.

Kyle: Well, I built it and then I showed it to Michael Seibel and a bunch of other people. And what I learned then was that everyone uses email in a slightly different way. And the reason Gmail is successful is it's like a Swiss Army Knife. It's very un-opinionated about how you should do email. And so everyone has their own filters, and some people are inbox zero, some people aren't, and the second you try to get to your opponent, you kind of piss off everyone. I don't know, I was overwhelmed with what it would take to get lots of users down that path. And I also realized that Google could just fix search and then I'd be in a bad spot, which definitely did. I looked on for some other ideas, and then one day I was just thinking about it. I think someone reached out to me that reminded me of working on self-driving cars, and I was like, "Man, it was one of the most fun things that I worked on in college."

Jessica: The DARPA, we're coming back to the DARPA Grand Show.

Kyle: We had a pickup truck. We made it drive by wire and put some sensors on. It didn't work super well, but it was fun. And I saw what Google was doing at the time with their self-driving car research project, which had spent a rumored \$100 million and all these great engineers working on it, but it seemed very academic. And I looked at the technology and the state-of-the-art, where sensors were, where software was. I was like, "Okay, maybe it's time for the lean startup approach applied to this, rather than the Google approach of treating it as a research project." And I thought about that for a while and identified what I thought would be the minimum viable product in self-driving. And once I thought that through, it really just clicked and I was like, "Oh, this would be great. I could work on this for a long time. It'll be fun. If it works, it'll be huge. Let's just go." And then the next day, once I had that epiphany, the next day I left Twitch and literally incorporated Cruise the next step.

Carolynn: You didn't get any downtime. You really just went from one to the next?

Kyle: Yeah, I think the one thing I could say is frankly, I wasn't operating at 100% for the last six months. My head was definitely in multiple places, so there's a pseudo transition in there.

Jessica: One of my questions that I had down was, were there things you learned from Justin.tv that made you more confident that you could make self-driving cars work? Do you think you already answered that? I'm curious.

Kyle: We had this attitude inside of Justin.tv that whatever the technical problem is, we'll figure it out and solve it. Whether it's bandwidth, or live-streaming, or building a new feature that's going to retain users. I think we were exceptional at just relentlessly pursuing whatever problem is in finding a solution. And so maybe some of that momentum or inertia had me think that even though Google was struggling to do this with their infinite resources and engineers, that somehow if I made the problem small enough, we'd be able to solve it. And I don't remember having much self-doubt about that. I think the biggest source of self-doubt or imposter syndrome was really just everyone I talked to said I was crazy. "This is never going to work and you should do something easier or you should just go work at Google if you like self-driving cars." That was the biggest source of doubt, not my own belief on whether I could build something.

Jessica: Interesting.

Carolynn: Just a lot of naysayers.

Jessica: Even in your world in San Francisco, people were still naysayers?

Kyle: Yeah, I mean, if you go too far outside of the comfort zone, you still face, people just look at you sideways and they won't tell you that they think you're crazy, but you can read it on their faces. I got a lot of that. Even after, Twitch was fairly successful at the time. It hadn't been acquired yet, but I remember going out and trying to raise money and for close friends that were doing angel investing, I could get some of those checks. There's almost like a handshake type thing in Silicon Valley where a lot of founders who are successful and friends just blanket invest in each other's companies. I got a little bit of money there, but it was really hard. I actually made 120 investor pitches over the first nine months of Cruise.

Carolynn: This is before doing YC?

Kyle: Before, during, and after to basically raise the money needed.

Jessica: How many was that?

Kyle: 120.

Jessica: 120. I swear I'm going to somehow package this information, because it's so fascinating to me how many "nos" very smart and talented people have gotten and they just keep going. You get so many nos before you get a yes.

Carolynn: Going back to something Kyle said, just from my very naive perspective, I was so used to YC funding software, so even I was like, "Cars, okay." I had the exact same reaction, I think that most of the people you knew did where it's like it felt very moonshot. Yeah, I was a doubter too. Sorry, Kyle.

Kyle: No, it's okay. Well, YC actually wasn't set up for companies that wouldn't have customers and users and revenue growth by demo day. I had these conversations like, "Well, what are you going to demo? You don't have customers, you don't have a product mature enough yet? It's like three months in for a significant development."

Jessica: What did you demo?

Kyle: Well, I mean, we work backwards from what is possible to show on this timeline. And we were able to get the initial products as a retrofitted Audi vehicle. We had a button and some cameras and sensors in there, and you could push a button and it would lane keep on the highway. And again, this is 2000, I guess 2013. Nothing happens-

Jessica: Winter 2014 you went through YC, so it would have been in March 2014?

Kyle: Right, right, right. You couldn't get a Tesla with this. There were some Lane Keeping Assists from various car companies, but it basically didn't work. This would actually keep you in the lane and drive you the whole way on Highway 101. And so I did that down to YC and that was the demo. But the pitch was very unique, I think compared to most YC companies at the time. It was basically attacking all the things that people had said in their no's from all the investor pitches I had done. And it was like, "Why are you going to succeed if Google is still struggling with this?" was the number one. And so I had this whole, "Small companies beat big companies and Xerox PARC analogies," and all this kind of thing in the demo day pitch.

Jessica: Wow.

Kyle: And then another one was like, "We're going to start small and sell this direct to consumer as opposed to needing to manufacture cars, which is a big problem." And then the other one was just a proof point that this small group of people can actually build stuff. And here's the proof, it drives on the highway, and granted it's not a fully baked product yet, but you could see very rapid technical progress in a short period of time. And so that's a reason to believe.

Jessica: Wow, a series of rebuttals in the demo day pitch, that is unique. I don't remember how investors responded, but I feel like I remember it was positively.

Kyle: My investment cycle, I've made a bunch of pitches before YC and got some checks from friends, and then it was basically all no's after that, or very, very small checks. Some people who normally invest half a million dollars or a million will say, "I'll give you 10,000 or 25,000 because I'm curious about this." And so I just had to scrape the bottom of the barrel continuously. And even after YC it started to get a little more money, but it was still a lot of ... It was better at that point. My yes ratio had gone up, because I think

probably by that time I was at least 50 or 60 pitches in. And you do anything that many times, just like they say in YC, you practice your pitch until it is impossible to forget.

And I had done this pitch and adapted it, and after getting so many nos and so much feedback. You kind of converge on a pitch that works. And for the last, I don't know, 30 pitches that I made, I feel like almost all of them were yeses. I got to this point where I had crafted the pitch and there had been some progress and I could just nail it, but it did take a crazy amount of meetings and pitches to raise money for this kind of idea, just because it was ambitious and I didn't have enough of a track record for people to just blindly invest large sums of money.

Jessica: I'm really shocked that it was that hard and you had to scrape together all this money.

Kyle: Yeah, well, I mean it's expensive too. That's another thing. Hardware stuff intends to cost a lot more. There's a lot more capital investment. And so the combination of higher capital needs plus a very ambitious goal and long timelines was not what most of the investors I was talking to were familiar with, not what YC was familiar with at the time, it's much better today. And so it was just that everyone was a little outside their comfort zone. And I think that was showing.

Carolynn: You talked about how much it was great to have the camaraderie of the Justin.tv folks and how that combated loneliness, but it's fair to say that you did Cruise all by yourself, right? You were a solo founder?

Kyle: Yeah. I mean, I tried to heed the advice of find a co-founder. I tried to, it didn't work. It's really hard to do that on short notice, if you don't already have someone you've known for a long time. And so I brought in two engineers from MIC that I had connections to and just hired them. I was like, "Well, I haven't found a co-founder that is going to stick here, so let's just proceed," basically was my attitude. It's like, "I spent long enough looking." Then a few months after that, Dan Khan, who was Justin Khan's brother, I talked to him and I was like, "Oh, he's perfect for this." And then I made him a co-founder at that point in time. I think that was maybe three or four months in. Really it was just that pure solo founder thing for the first three or four months. But then after that, I had a similar form of that camaraderie and the thing people get, emotional support, all the other things in addition to the actual doing a lot of work that you would expect from having a co-founder.

Jessica: Obviously, you met Daniel through Justin, but what was Daniel doing at the time? Was he in San Francisco?

Kyle: I don't remember exactly what he was doing at the time, but his trajectory was, he was an intern at Justin.tv around the time he was still in college. He was a year behind me, I think. And then he and Justin had started Exec, you remember that company we hired-

Jessica: Oh yes.

Kyle: Hire someone for \$25 an hour and they'll do whatever you want. And then that pivoted to a cleaning company, weirdly house cleaning. Because everyone's like, "Can you just clean my house?" And they say they pivoted into that. And then I think there was one other startup that he worked on between that and Cruise, but he had grown very, very quickly in his own journey and becoming an entrepreneur and finding creative ways to solve really grungy, sometimes unsexy problems. And so I had an appreciation for that. And frankly, the whole Khan family is very special. There's a lot of grit and integrity and good things that I saw in Justin that I also saw in Dan.

Jessica: You're with Daniel, you've raised some money. What turned out, sometimes sort of the nastiest problems with hard problems turn out to be not what you expected and not what you predicted. What was one of the toughest things about the self-driving car technology?

Kyle: We started off with this retrofit play, but from day one the goal was this sort of one-two business model, which may not be a great idea, but we were going to do this retrofit strategy, get some revenue and use that to fund the development of fully autonomous cars. Not a retrofit, semi-autonomous thing. We wanted to build fully autonomous cars. And what we learned after trying to build this retrofit thing is we really had two big problems in front of us in addition to actually making the thing work. We felt good about that, spending more time and money on it. The first was that it was a very complicated legal and liability regime between a human driver, the person doing the retrofit and the car maker. And it's a highly regulated industry, and so possibly we could have found a path forward there, but it's very complex, like what happens if something goes wrong, who's at fault? And especially doing that without the explicit partnership with the car makers, which we didn't think we would be able to get at that stage.

The second was we put up a pre-order page for this product, just to see what would happen and who would pay for it. And then part of that was like you didn't have to put down money, but you had to tell us what kind of car you were driving. And we had hoped that if everyone put in what kind of car they drove, we would see some patterns from our potential customers. We'd hope they'd all drive BMW 3 Series, or a Honda Civic, or something so that we'd only have to make this retrofit work on one or two kinds of cars to get lots of customers.

And what we learned was that there were no patterns. Very few people drove the same car as other people. And so for us to get a significant number of customers, we'd have to configure the system for lots and lots of vehicles, which just adds an entire new dimension of complexity and work. Because at the time, we actually had motors that were physically turning the steering wheel. And so if you can imagine trying to build that contraption and adapt it to every different car steering wheel and configuration would have been a mountain of work. And so we realized that we were maybe at the end of the road for that. And then the big pivot moment for us was realizing that Uber and Lyft were exploding in popularity. This is, I don't know, 2014, 2015. And their business models are fundamentally flawed. And we've seen that because just I think over a decade later, Uber is just now turning a profit for the first time, and that's because all the money basically has to go to the driver for this to work and at least very little left for the company.

And so obviously, if you can not have the driver and have a self-driving car instead, that business just explodes. And so that seemed very lucrative. And then we were able to convince Nabeel Hyatt at Spark Capital to do this weird thing where he was going to fund our series A simultaneously with a pivot. We were going to shut down the old idea to do retrofits and do a hard pivot into working fully on full self-driving cars to go after the robotaxi market. And he was actually investing in that new thing, not the old thing. And kudos to him for betting on us and the team, but I think we could only pull that off, because we were able to show him, "Look what we built in this very short period of time, this functional thing. We're going to take all these brains and all this talent and now put it to this problem and you should expect to see the same kind of success there." And he took a bet on us.

Jessica: How much did you raise in that series A?

Kyle: I think it was about 12 million.

Jessica: I'm impressed that you could pull that off. I mean, VCs don't usually fund for the pivot.

Kyle: Yeah, it's unusual. I don't know if that was Nabeel being early in his VC career and making a dumb move and getting lucky, or he really saw something in us. Either way, he's excellent and gone off to be one of the best VCs out there. So, I think he must've seen something in us.

Jessica: Wow, okay. And you were in San Francisco, Ed Lee was still the mayor, right?

Kyle: That's right.

Jessica: There's two things I want to ask you about. One just for the listeners who might not understand the safety levels of self-driving cars. I'd love for you to explain how it actually is much safer, and that's why you're doing it and that's why Cruise is so successful. And can you explain that a little bit?

Kyle: Yeah, I mean, that's one of the big questions in the industry and one we struggled with. Because unlike building a plane or a regular passenger vehicle, or developing a new drug, there is no regulatory framework that tells you, "Here's the checklist of things you have to do, or here's how good the product needs to be before you launch." I wish the federal government had foreseen this problem and jumped on it sooner, but they hadn't and we wanted to build this now, so we couldn't wait until they do that. And so the big challenge, and it's a combination of an ethical and business and moral kind of thing, is essentially how good does this need to be before you launch it? And we all know the answer can't be perfect, because no typical system is going to have zero flaws. And we also know that status quo is very bad with 30 to 40,000 people dying in car accidents per year.

And so, it would reason that somewhere between where we are today and perfection is when you should launch. And so to establish that we spent years researching how good human drivers are in the city of San Francisco where we wanted to launch first. And so

there were hundreds of cars instrumented with sensors that were driven around by Lyft drivers. There were millions of miles of data that we were able to access from GM's OnStar networks of people driving around GPS tracks from people driving around essentially, they were anonymized. And then some academic sources of data that we were able to mix together and produce this pretty rigorous estimate in partnership with the University of Michigan for how safe human drivers are. And what we looked at is how often did they have a collision and for each collision, how severe was that collision? And we had basically a ranking system for that.

And internally our North Star was that basically under no circumstance should we deploy this product if we ever are below or fall below what that human status quo benchmark is, that's the minimum. And of course, we aim to significantly overshoot that. And anyone who knows statistics knows if you want to say you're better than some benchmark, you need statistical significance, which means you need to be way better than your benchmark, or drive a whole lot of miles to prove that you're just a small margin above that benchmark. And so we ended up doing that. It took us, it was like a seven or eight year grind to actually get the technology from proof of concept to the point where it beat our hurdle for human performance by a significant margin to the point where we could initially launch.

And even once we did that, there's still a risk of unknown unknowns. And so we had an envelope expansion approach where we'd start with a closed course, no people in the car, no people around the car and operate that, we called it CUDL, continuous unmanned driving loop. And we just drive the car in circles for a really long time and gain confidence that the system performed as expected. And then we would do a more complex test track. And then our first deployment on public roads was in a remote part of San Francisco in the middle of the night. And so even though we felt like we were way above that safety bar, we still started in an area just to minimize any risk. And then once we gained some confidence there and collected some data, we'd expand the hours in the region gradually to the point where we were able to operate in the whole city.

And so it was a very measured, cautious step-by-step approach. And anytime we discovered something, we would either immediately hit the brakes on our operations if it was a severe thing, which did happen on a rare occasion, or basically triage issues and make sure we were always working on the thing that was going to improve safety the most at any given time. And that's how we got this, I think, pretty rapid performance improvement over time.

Jessica: In retrospect, was it worth doing it in San Francisco? Because when I brought up Ed Lee, I mean, I feel like that was a different world in San Francisco and now it's dysfunctionally against tech. But you're in San Francisco, so you have the best engineers. Should you have launched in a different city, or does it not matter?

Kyle: Well, we chose San Francisco for a few reasons. The first was we're all based here, and so we can dog food the product and use it every day, which is really helpful just in general. The second was, San Francisco is a hard city to drive in. It's got hills and fog and lots of construction and weird traffic patterns and uni buses and all this stuff. If you want to improve a technical system most rapidly, I think you expose it to the toughest

conditions. You learn more quickly from that. You'd only have to drive a mile or less in San Francisco to see something really complex and challenging. And so that kept us from having any sort of false hopes about where we stood. We knew exactly where we stood by driving in San Francisco.

And then the third is that like a robotaxi business model kind of depends on density. You need a density of demand and riders, and you need a density of vehicles in order to keep the wait time short and to keep the vehicles highly utilized. We had to be in a big city to make that work. In retrospect, I think there would have been better cities to launch in. There's this weird tech lash undertone going on in San Francisco, and I've seen many other [inaudible 00:44:42] across other industries bump into this. There are other places that truly want to be seen as innovative and more willing to work more with tech companies versus treating them potentially as the enemy, depending on who you're talking to. And I think we underestimated just how significant that could be, that pushback. Despite the fact that from our perspective this very logical argument of these things are having a huge positive impact on road safety overall compared to humans. They're all electric, they are way better for the accessibility and disability community.

And we were even offering rides for free for college students and other things and delivering meals for the homeless. And literally all this stuff. We're doing all these things that were like, this is clearly positive for this city, but those logical arguments don't hold up against whatever a city or various government officials' priorities are. And that was surprising to me. It was a huge lesson learned.

Jessica: Could never do enough. I remember when I took a ride in Cruise in one of the cars, I think it was like you either had been acquired recently by GM, it was around that time, and Paul and I visited your office in San Francisco and we went for a ride. Would that have been the right time frame?

Kyle: Yeah, I think so. I think it's much different to get in the car with no driver and the more recent stuff compared to a long time ago. But we were always big on trying to make the vehicles usable and testable because that was the only way we could make sure that we were on the right track. You have all these metrics and things, but the metrics tell one story and then when you get in the car, there's always a different feel and vibe to it. And for me, there was always a process of trying to refine what we're measuring to encapsulate all the stuff that I was annoyed by when I go on a ride. And actually, it turned out to be really hard to come up with a metric for driving that doesn't suck, basically. But we were constantly trying to do that.

Jessica: And so you got acquired by GM?

Kyle: Yeah, initially we were rolled in with our org chart intact, but seen as an R&D unit inside of GM.

Jessica: Yes.

Kyle: That was hard because we wanted to launch products and do things. And it's hard to do that when we're several levels deep in a corporate structure. That knowledge plus the idea that this is going to be capital intensive led GM to ultimately spin out Cruise as a separate company again in 2018. And that was simultaneous with taking on a big investment from SoftBank and from Honda and agreeing to build a new vehicle together with Honda. And then also Dan Ammann, who was the president of GM at the time, jumped over and became CEO. Took my job at that time. And I think frankly they wanted some adult supervision for this massive capital investment that they were doing. And I happen to really like Dan as a person in working with them, so I was supportive of that move, but that made the company we were kind of acquired and then spun back out a few years later.

Carolynn: Can I ask, back in 2016 when GM approached you, was your attitude ... You've been this scrappy, very scrappy startup for a while, you've been doing it, I'm sure there was a lot that went into that decision to go ahead and take the offer, but was it because you knew you couldn't get further without a huge infusion of cash?

Kyle: Yeah, we knew it was going to be very expensive, regardless, and thought that Dan Khan and I had many discussions about fundraising and we were preparing to go out and raise a round at the time that we decided to get acquired instead or to sell the company. It was like we saw a path of being independent. There was a huge cloud of ambiguity around funding and also scale, how you either build lots of cars to be robotaxis or retrofit them and what that looks like. And the pitch from GM, and actually Dan Ammann, who was sitting on the other side of the table from us at the time, was that, "Maybe you can go off and do this independently, but with the scale and resources and manufacturing capacity of North America's largest automaker, don't you think you could do that faster?"

And we had to ask ourselves, "What's more important here? Is it an ego thing about staying independent, or do we want this mission to have a higher probability of success?" And the financial terms were good, they were healthy for us. It was a good return for everyone, and it was a premium to the valuation we thought we would've gotten in the next funding round anyway. That kind of checked out. And we were so used to selling everyone on how important this mission is, and when it came down to it, I think we had to pick the choice that we thought at the time would increase the probability of success. That seems more important than anything else in the end.

Carolynn: How did they hear about you? Was it just you'd gotten enough press and enough buzz that you caught their attention?

Kyle: A lot of these big auto companies or other companies seem to have little scout groups, venture groups that GM had a venture arm. They go around and knock on all the startup doors just to see if there's anything coming. And then they would do a thing where they'd try to lock up a deal and have exclusive access to that technology or whatever the company was building so that their competitors couldn't get it in exchange for an investment. It was a good strategy and they tried to pull that on us and we were like, "If you want exclusive access, you have to buy the company and the company is not for sale." That was our initial answer. Ultimately that changed, but they had people out

there and tendrils out there looking for all sorts of cool technology and things that could help their business.

Carolynn: Yeah.

Jessica: What did working when you were working within GM, what did that experience teach you about the advantages of working at a startup?

Kyle: It was really fascinating for me just to learn how large companies work. You think about a company, it's General Motors, but it's really like insert any, probably company, in the Fortune 500, that's been around longer than 20 years. They have so much institutional knowledge that's built up over time. And it ends up being codified in process and then surrounded by bureaucracy so that that process is always followed and not deviated from. And so that tends to be really good at making sure their core business becomes more efficient, more successful, more profitable. But the people involved in this giant machine get promoted essentially for not making mistakes, not for taking risks and failing, but for not taking mistakes. And so the culture then becomes risk aversion. And after this continues for a certain period of time, no one who works at the company knows why all the processes exist or remembers the problems that led to those processes being created or knows why the bureaucracy operates the way it does.

And so it becomes hard then if you don't understand the reasons for things to accept any deviation from that process or status quo. We come in, a startup, and we're just like, "We just need to do whatever the most important thing is next to get the product working. And rules. Maybe if they apply for a good reason we can debate that and have a logical argument about whether that's appropriate here, but we don't really care. We just need to get things done." And that is just hitting a brick wall. When you try to have those discussions with people who are well-intentioned, but focused on not making mistakes and keeping this process running, not on sticking their neck out and taking your risk on deviating from the rule and then they might get fired.

Carolynn: Yeah.

Jessica: Interesting.

Carolynn: So, you said that Dan came in and is now the CEO of the spun out company, right?

Jessica: Yes.

Carolynn: Did you leave at that? Is that when you left, or did you just take a different role?

Kyle: Yeah, I just love working on this stuff and still wanted to make the self-driving cars work. So, I took a step to the side and had a whole long list of titles that maybe were there to make me feel better or something. But CTO, president, co-founder, this big, very elite stack of titles, but I feel like we mostly ran the company together, but it was clear that he was, he is a CEO got the final say on things. But on the other hand, once he stepped into the startup we had created, it was a few hundred people at the time, he very clearly

could see that there's a different energy, a different DNA here, and really needed to lean on Dan Khan and myself to figure out how to navigate this new environment and do it well. And to his credit, he adapted pretty well and figured out what it took to be successful in that very different environment from the corporate environment at GM.

Jessica: As startups get bigger, the founders have to grow into this management role and hire C-level executives, and that's very challenging for a lot of technical founders, but I feel like you were good at it, right? Do you feel like that was a challenge for you?

Kyle: We found a way to work around any hiring mistakes we made and ultimately move the company forward, but we definitely made some mistakes. I felt like hiring executives is really hard, especially if you've never done it before, because they interview so well. They come across just like ... They championed their last company through battles and won wars, and they carried the entire weight of it on their shoulders, and they're alive to tell the tale and they know everyone in the industry and will immediately recruit 50 of the world's best X to show up at your office the next day. The stories are remarkably similar. And so teasing through that and finding people that are actually going to be successful in this particular company for this particular stage is really hard. I feel like we made a ton of mistakes. I think we were decent about correcting those fairly quickly when we did, but it took me a long time and a lot of mistakes to feel like I was a better judge of these types of candidates. It's just really hard.

Jessica: Is that the key sort of correcting the mistakes quickly versus you'll never really be able to see through and hire the exact right people?

Kyle: Yeah, I mean look, it's pretty devastating if you hire a C-level person and introduce them to the company and then they leave for any reason, right? It's hard, because change at the top is very disruptive for a larger company or at least signals some sort of distress or problems. I think you got to correct things quickly. It's still very painful whether you do it quickly or wait a long time. I guess from that perspective, you might as well just rip the Band-Aid off.

Jessica: Yeah. All right, we're running out of time. I have a couple random questions that I want to get to. I believe you introduced Drew Houston and Arash of Dropbox. Is that true? Tell us the story.

Kyle: Arash and I both grew up in Kansas and got accepted into MIT in the same year. And so there was this little get together for all the new incoming students who lived in Kansas, all four of us. And so we showed up to this thing and I met Arash there, and then we stayed in touch in college. I just remember him being super smart. We would work together on our calculus homework and he would just blaze through it and I'd be asking him all sorts of questions and everything. I remember him as smart and also a little edgy. He had some side projects that he was hacking on and some schemes to make money on various things. And I was like, there's something interesting there, in him.

And I met Drew at the entrepreneurship club. He was at MIT, he was a couple of years older than me. And we were just taking turns pitching each other on stupid business

ideas basically, and became acquaintances and friends through that. And then he told me about Dropbox and said he needed a co-founder. I was just like, "You should talk to Arash." And I think they met the next day and agreed to be co-founders an hour later. It was super fast.

Jessica: Wow.

Kyle: It was a very serendipitous thing and I'm really glad that worked out because I think they're both amazing people.

Jessica: That really worked out. I don't think...did Arash then have to drop out of college?

Kyle: He did. I think he kind of did what I did. I was maybe six months or a year ahead on this identical journey. And actually, I remember his dad called my dad, he told his dad, "Kyle dropped out and he's doing fine. Look at this Justin.tv thing." And I don't know how convincing that was, but I remember our dads talked and I don't know what happened in that conversation because my parents weren't too happy about it either at the time. But eventually Arash ended up dropping out, I think, and doing Dropbox full time.

Jessica: The Kansas Dads Club. Oh my God, I'm so glad I asked that question. Just to learn that your dads talked to each other about this whole dropping out business.

Kyle: I'm surprised I didn't inspire or force us both to go back to school, but we managed through that somehow.

Jessica: The other question I wanted to ask you was, you're running of the marathons on a vegan diet on every continent. Tell us about that?

Kyle: Yeah, that was kind of wacky. We were at this point in Cruise where the seven-year grind, where the only thing we're doing is measuring our performance, safety and other elements of performance, and it just needs to get better, several orders of magnitude. It's like a long grind. Sometimes we make a breakthrough, sometimes there's a setback. And I felt like I needed something in my life where I knew I could just put in the time and get the results. And so I was like, I like distance running. I read about this crazy race where these people would take you around and they would run a marathon on each continent each day for seven days. And I was like, "Oh, this is interesting." But I was looking at it and the person who held a record for that, he had done it in five and a half days, instead of seven days-

Jessica: Wait, ran a marathon every day for seven days for five and a half days?

Kyle: Someone else had done it outside of this and done it in five and a half days, a marathon on every continent.

Jessica: Oh my God. Okay.

Kyle: I was like, "Oh, interesting. What's the theoretical fastest time?" And so I wrote some software to calculate, it's kind of the traveling salesman problem. What's the shortest possible route, taking into account different plane speeds and winds and everything else, that theoretical minimum to touch all seven continents, landing in airports that you can actually go through customs on. And it turned out to be 29 or 30 hours, and here the record is five and a half days. And so I'm like, "I think this is possible. I think if you optimize the route and do the shortest possible path, even if you're not the best runner in the world, which I'm not, I think you could beat this guy's record."

And so once I had that idea in my head, I couldn't get it out, figured out this route, figured out the logistics, had a great team put together to do this. It's been 18 months training to basically get myself into the shape where I could do this and not destroy my body. And I was doing it all on a plant-based diet. At the time I was really interested in blowing up this notion that you needed animal protein to do crazy athletic things. Ended up doing it. It started in Antarctica and then had crazy logistics and flights set up. I'd basically land in an airport in my running gear. The second I got out of the airport, I literally start running a marathon. I had a course chart out, and then the second I finished, I went back, you go to the airport again and go back on a plane and fly to the next one. And I would try to eat, sleep and recover on each flight.

Carolynn: That's amazing.

Kyle: My world record now is 81 hours, instead of the five days, there's like 120 hours. I think that should stick around for a while.

Carolynn: Yeah, I was going to say, now you have the record to beat.

Kyle: Yes. For the crazy ones out there, please try.

Carolynn: I didn't think you could run a marathon on Antarctica. I had no idea that was a thing.

Kyle: You got to land a plane on ice, which is a little unnerving to get there.

Jessica: Was that the hardest continent to do the marathon on?

Kyle: I think the hardest for me was actually, technically it was North America. It was like Panama City, just past the tip of South America. And the problem was, which I didn't really fully plan for, is that your body takes some time to adapt from cold weather to hot weather. When you're in cold weather, like Antarctica, your skin adapts to hold in moisture and do various things. And then it takes a little while to un-adapt from that. But by the time I got to Panama City, it was like 90 degrees and 90% humidity, and my body was not adapted to hot weather yet. And so I was going hypothermic there, which is not from being too cold, but from overheating. And so I had to strap bags of ice to my chest to keep running. I brought a doctor with me and he's like, "This is normally where I tell you to stop, but if you wanted to keep going, you need to find a way to cool yourself down."

Jessica: Was anyone documenting this?

Carolynn: Why is there not a documentary or a book about this already, or something?

Kyle: We got some film from it. I mean, maybe one of these days.

Jessica: Oh my God, get someone to put this together, because I'll let you go. But there's so many more questions that I have. What was the moment? Was that strapping the ice in Panama to your chest, was that the lowest moment of the whole process?

Kyle: Yeah, that was it. I mean, I felt like I had done the training. I knew I had the necessary fitness, but my body was just acting differently. It was shutting down and I was weirded out by that. And that was the moment I wanted to give up, because all my preparation was not helping me at all in that moment. That was hard. But I got to say, doing something like that in the middle of running a startup, after that any kind of problem that I would face, running the company, hiring someone, a weird setback, I felt like it now fell somewhere between the highs and lows I had experienced elsewhere. And so I felt like it just took the edge off a lot of things, which is really helpful when you're operating a tough startup in a tough space and doing it for a very long time. I don't know if I'd recommend doing that for other entrepreneurs, but having something where you push yourself to the limit and expand your dynamic range of stress that you can handle, I think helps a lot.

Jessica: Wow. And now, Kyle, you are a free radical exploring ideas and things?

Kyle: Yeah, happily unemployed. Probably not for long. I think there'll be another company in my future, but I'm trying to figure out what that is. What's the next important problem to spend a decade on?

Jessica: Well, here's a question, I won't ask you what you're doing, because I don't know if you probably want to share that, but you are now, are you 38 or 39?

Kyle: 38.

Jessica: 38. You're kind of on the older side for being a founder. You're not old, but for being a founder, I mean, you started off at age 20, so do you have any opinions on being a young founder versus an old founder? Is there anything that young founders have that are an advantage over older founders?

Kyle: Well, we'll see. I mean, you're the first person who's told me I'm an old founder, so. Moderately offended by that. But no, I think the big difference is kids and family stuff usually looks different in your 30s than in your early 20s straight out of college. And so I'm still wrapping my head around what running a startup looks like in that mode. Because it's not what it looks like when I was starting Justin.tv. It's not what it looked like when I was starting Cruise. But on the other hand, I see lots of people doing that successfully. And so I think part of it is just being real with myself and my family and the company, what that looks like. And so, I'll probably talk to a bunch of other people in this

situation, but I can imagine it's not staying up till four in the morning, overloaded on caffeine, hacking away. I don't think it'll look quite that way this time around.

Jessica: Yeah, that is a good answer. I'm very curious about this subject. Sometimes I think that the younger founders are ignorant and in a way ignorance is bliss when you attack hard problems. And maybe when you're older you don't quite have that same ignorance. You've been around the block.

Kyle: Yeah. You have to, I think, replace the ignorance with a little bit of optimism and maybe even overconfidence. You're aware of all the problems, but you decide to do it anyway. Whereas when you're young and naive you're just not even aware of the problems. It's a slight difference, but I think you can do it either way.

Jessica: Yeah. Awesome. Well, this was so much fun. Thank you for your time. It was such a fascinating conversation. I loved it.

Kyle: Good to see you both. Thanks for having me.

Jessica: Nice to see you too. And enjoy your momentary unemployment, Kyle.

Kyle: Will do. Thank you.

Jessica: All right.

Carolynn: Good bye.

Jessica: Bye. Carolynn, I loved catching up with Kyle.

Carolynn: It was great. And we didn't even have enough time. There were so many other questions and other stuff to talk about with him.

Jessica: I know. And I was so shocked. There were things I learned during this conversation.

Carolynn: The marathon thing, I guess I remember someone mentioning that.

Jessica: You have to remember that. Do you not remember that?

Carolynn: I mean, how did I forget that? But yeah, I guess I didn't fully absorb it, but it's astonishing. And I really would watch that documentary. When we were talking about San Francisco, I kind of got sad, because I would see the footage of the Cruise cars getting randomly attacked by randos who just wanted to break the windows and set it on fire. And I could be conflating it, but that just is heartbreaking. It's like, here's this thing that is doing all this good and-

Jessica: And is inevitable.

Carolynn: Yeah.

Jessica: I don't know what the timeframe will be. I don't know if it's 10 years, 100 years or whatever, but there will be self-driving cars and we will be ... Or our kids will be riding in them.

Carolynn: Exactly.

Jessica: It's inevitable.

Carolynn: Yeah, that's such a good thing for humanity. We all are terrible drivers, especially with our cell phones that distract us.

Jessica: And 30 to 40,000 unnecessary deaths every year. I mean, I don't know why people are fighting against this. They should be embracing it.

Carolynn: I don't either. I don't either.

Jessica: But he's just an incredible founder, an incredible technical person, and an incredible human being. How classy is he?

Carolynn: He's incredibly classy.

Jessica: And diplomatic.

Carolynn: Yeah, he's classy.

Jessica: I have to admit, there were some questions I wanted to ask him that I didn't, because I could tell he's just ...

Carolynn: Too classy to talk about it.

Jessica: Too classy to talk about it.

Carolynn: Yeah, exactly.

Jessica: Whereas I wanted to, but I didn't.

Carolynn: The other thing that I...so I think it's one of the best takeaways, it's a little generic just to broaden it like this. But when he was sitting there thinking about, "What do I want to do next?" And he thought back to the DARPA challenge. And so making the connection of, "When did I have a great time surrounded by super smart people doing super hard things?" I think that's a good nugget about startups just generally and how founders operate, so I really actually liked that a lot. And I wonder, we could all think about that, like, "When were we happiest doing something hard and cool like that?"

Jessica: Yeah. Y Combinator for me, for sure.

Carolynn: Yeah, exactly. Exactly.

Jessica: But you just know also, I mean, this is taping well in advance of when it's going to air. I wouldn't be surprised when this podcast airs if he has started his new thing. 100% I bet it will be just as ambitious as Cruise.

Carolynn: Yes. Yes. And just as technically difficult.

Jessica: So many people could just rest on their laurels and think, "Okay, I'm going to find some hobbies and just hang out with my family and enjoy life."

Carolynn: I mean, in many decades time we should look at Kyle and say, "Is he the guy who has done the most startups?" If he keeps doing them successively because he keeps wanting to do new challenges and stuff, maybe he's, because already has quite a few under his belt.

Jessica: I know. I'm so glad we got him on the show, and I think it's going to be a great episode.

Carolynn: I do too. I'm looking forward to it.

Jessica: All right, I will talk to you soon, Carolynn.

Carolynn: Bye.

Jessica: Bye.